PROPRIETARY & CONFIDENTIAL — LWANDILE ENGINEERING PROJECTS & INCLUSIVE ENTREPRENEURIAL ACADEMY

SCAPUSH PRECISION STANDARD

SECTION 04 — LAND & SITE STRATEGY

Sovereign Bio-Solar-Thermal Campus | Master Design Specification

Doc Ref: 04_land_and_site_strategy.html | Status: APPROVED BASELINE | Governance: Scapush Precision | JET Impact Aligned

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1. Site Acquisition, Boundary & Tenure Baseline

LISTEN SECTION

The Land & Site Strategy establishes the physical footprint, legal tenure, and environmental boundaries for the Sovereign Bio-Solar-Thermal Campus. To satisfy JET PMU risk mitigation standards, the site operates under an unencumbered leasehold model preserving 100% community title sovereignty.

Ref Code	Site Parameter	Specification Boundary & Strategic Value
LS-001	Land Area Footprint	20,000 m^2 (2 Hectares) minimum cleared build site with 565m perimeter.
LS-002	Tenure & Legal Model	Long-term Leasehold Model (Zero property title transfer or debt pledging).
LS-003	Primary Site Purpose	Integrated Off-Grid Food-Energy-Water (FEW) Demonstration Campus.
LS-013	Expansion Capacity Reserve	Dedicated spatial reserve protected from permanent construction in Phase 1.

2. Environmental Orientation & Wind Mitigation Strategy

LISTEN SECTION

Spatial zoning follows strict microclimate and environmental vector planning. Production zones carrying elevated biological and olfactory risk are situated downwind of human-centric administrative and educational zones.

Prevailing Wind Strategy

The site shall be developed with due consideration for prevailing wind conditions. Production zones with higher biological risk shall be positioned downwind of administration, academy, and visitor zones wherever practical. The relative positioning of poultry, swine, biodigester, and composting facilities shall seek to minimize:

- Pathogen transmission and aerosolized disease vector migration
- Dust and airborne particulate migration
- Odour transfer to public and academic facilities
- Surrounding air quality and community amenity impacts

3. Security Layer Separation & Access Control

LISTEN SECTION

To ensure biosecurity compliance and public safety, the campus enforces a strict three-level security perimeter architecture separating public community zones from controlled energy and livestock systems.

Three-Level Security Architecture (LS-016)

Security Zone	Access Classification	Permitted Infrastructure & Personnel Scope
Public Zone	Unrestricted / Community Access	Administration, Inclusive Entrepreneurship Academy, Accessibility Hub, EV Charging.
Controlled Zone	Permit Required / Operations	General Operational Facilities, Storage Warehouses, Water Sovereignty Centre.
Restricted Zone	Biosecure / High Vulnerability	Livestock Pods (Poultry & Swine), Biogas Energy Centre, Solar & BESS Compound.

4. Core Production & Energy Infrastructure Allocation

LISTEN SECTION

Physical spatial layout allocates distinct infrastructure modules designed around complete off-grid sovereignty and zero fossil fuel usage.

Agricultural Infrastructure Allocation

- **LS-004 (Barn 1):** Poultry Production Cluster (12m × 32m, 4 Biosecure Pods, 4,800 to 5,720 Broilers).
- **LS-005 (Barn 2):** Swine Nursery Cluster (12m × 32m, 8 Pens, 160 to 500 Weaners, Feedstock Role).
- **LS-011:** Water Sovereignty Centre (Borehole, UV Treatment, Elevated Storage & Gravity Distribution).

Energy Infrastructure Allocation

- **LS-006:** Permanent Off-Grid Campus Sovereignty.
- **LS-007:** Strict Diesel Prohibitive Clause (Zero Diesel Generation).
- **LS-008 & LS-012:** Integrated Solar PV Array + BESS Storage + Biogas CHP Energy Centre.

5. Utility, Data & Internal Transportation Networks

LISTEN SECTION

Reticulation networks are buried in dedicated, protected corridors running parallel to internal roadways to prevent operational disruption during maintenance or future phase expansions.

Utility & Data Corridors (LS-014 / LS-015)

LS-014 Utility Corridors:

Dedicated pathways accommodating electrical reticulation, water pipelines, and future expansion space without excavation disruptions.

LS-015 Digital Infrastructure Corridor:

- Fibre optic backbone provision
- Ethernet infrastructure for SCADA communications
- Wireless telemetry & ArIoT mesh networks
- Remote monitoring expansion pathways

Internal Access Road Network

An engineered internal road network accommodates specialized logistics vehicles:

- Feed delivery heavy vehicle access
- Veterinary & biosecurity vehicle access
- Maintenance & utility maintenance access
- Emergency vehicle priority routes
- Livestock logistics & off-take vehicle access

6. Social Beneficiation, Emergency Safety & Expansion Protection

LISTEN SECTION

Social & Educational Centers

- **LS-009 Accessibility Hub:** Universal step-free access, wheelchair charging, and DC EV charging bays.
- **LS-010 Entrepreneurship Academy:** Partnered facility led by Dr. Palesa Charlotte Felix-Faure for green enterprise training.

Emergency Muster Area & Expansion Protection

Emergency Muster Area:
A designated, accessible emergency assembly area established in the Public Security Zone for staff, trainees, and visitors during emergency incidents.

Expansion Capacity Protection:
Future expansion areas shall remain free from permanent construction during Phase 1 implementation unless approved through formal design review.

7. Land & Site Strategy Frozen Lock-In Register

LISTEN SECTION

Ref Code	Locked Core Parameter	Design Specification & Governance Basis
LS-001	Land Area	20,000 m^2 (2 Hectares) minimum secured footprint.
LS-002	Tenure Model	Long-term Leasehold Model (Sovereign community land preservation).
LS-003	Campus Nexus	Food-Energy-Water (FEW) Demonstration Ecosystem.
LS-004	Barn 1 (Poultry)	12m × 32m Barn | 4 Biosecure Pods | Broiler Production.
LS-005	Barn 2 (Swine)	12m × 32m Barn | 8 Pens | 160–500 Weaners (Feedstock Role).
LS-006	Off-Grid Strategy	100% Off-Grid Independent Power Operation.
LS-007	Prohibited Energy	Zero Diesel Generators (Strictly Excluded).
LS-008	Energy Generation	Solar PV Array + Battery Energy Storage (BESS) + Biogas CHP.
LS-009	Accessibility Hub	Universal Access, Disability Priority, Wheelchair/EV Charging.
LS-010	Skills Partner	Inclusive Entrepreneurship Academy (Dr. P. C. Felix-Faure).
LS-011	Water Centre	Water Sovereignty Centre (Borehole, Filtration, Gravity Distribution).
LS-012	Biogas Centre	Biogas Energy Centre (Anaerobic Digester + CHP Heat Recovery).
LS-013	Expansion Reserve	Future Expansion Capacity Reserve (Protected in Phase 1).
LS-014	Utility Corridors	Dedicated Underground Power & Water Reticulation Corridors.
LS-015	Data Corridors	Dedicated SCADA, Fibre & Telemetry Digital Corridors.
LS-016	Security Control	Three-Level Perimeter Security Separation (Public, Controlled, Restricted).

8. Approval & Authorization

LISTEN SECTION

DOCUMENT STATUS

STATUS: APPROVED BASELINE (REV 0.1) — LOCKED FOR JET PMU REVIEW

PROJECT STEWARD & TECHNICAL LEAD



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Sipho Wiseman Khuzwayo
Inclusive Legacy Architect / Principal Lead
Lwandile Engineering Projects (Pty) Ltd
Status: Signature Image Loaded

GOVERNANCE & SKILLS PARTNER

Click or drag & drop signature file
(JPG / PNG / SVG)

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Dr. Palesa Charlotte Felix-Faure
Founder & Academic Lead
Inclusive Entrepreneurial Academy
Status: Awaiting Signature File

Consortium Leadership

Sipho Wiseman Khuzwayo — Lead System Architect (Lwandile Engineering)
Dr. Palesa Charlotte Felix-Faure — Academic & Governance Lead (Inclusive Entrepreneurial Academy)

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Execution Entities & Standard

Lwandile Engineering Projects (Pty) Ltd
Inclusive Entrepreneurial Academy
Scapush Precision Execution Standard

Operational Footprint

Sovereign Bio-Solar-Thermal Campus, Mpumalanga, South Africa